A COUNTABLE GENERATION PLUS SPLITTING OF ELEMENTS INTO FREE SUMMANDS MAKES A MODULE FREE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Direct sum decompositions and free direct summands. Used throughout the counted proof (lines 20841-20853). Allows the module to be split off step by step into finite free pieces.
- The stated property applied to complements. Used at the induction step (lines 20849-20851). Produces the next finite free direct summand containing the image of the next generator.

Lemma 0.1. *Let M be a countably generated R-module with the following property: if $M = N \oplus N'$ with N' a finite free R-module, then any element of N is contained in a free direct summand of N. Then M is free.*

Proof. Let $x_1, x_2, \ldots$ be a countable set of generators for M. We inductively construct finite free direct summands $F_1, F_2, \ldots$ of M such that for all n we have that $F_1 \oplus \ldots \oplus F_n$ is a direct summand of M which contains $x_1, \ldots, x_n$. Namely, given $F_1, \ldots, F_n$ with the desired properties, write

$$M = F_1 \oplus \ldots \oplus F_n \oplus N$$

and let $x \in N$ be the image of x_{n+1}. Then we can find a free direct summand $F_{n+1} \subset N$ containing x by the assumption in the statement of the lemma. Of course

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we can replace F_{n+1} by a finite free direct summand of F_{n+1} and the induction step is complete. Then $M = \bigoplus_{i=1}^{\infty} F_i$ is free. $\square$